Roll Number | | | | | | | | | | | | |

NARULA INSTITUTE OF TECHNOLOGY
An Autonomous Institute under MAKAUT
2026
Even Semester - 2025-26
EC(DS)201 - BASIC ELECTRICAL AND ELECTRONICS ENGINEERING

TIME ALLOTTED: 3Hours **Maximum Marks : 70**

Instructions to the candidate:

Figures to the right indicate full marks.
Draw neat sketches and diagram wherever is necessary.
Candidates are required to give their answers in their own words as far as practicable

Part A
Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

1.	The energy gap between the valence band and conduction band is known as:	**(1)**	CO3	BL3	PO4
	(a) Bias Voltage (b) Barrier Potential (c) Forbidden Energy Gap (d) Breakdown Voltage				
2.	The output of an inverting amplifier is:	**(1)**	CO4	BL3	PO7
	(a) In phase with input (b) 90° out of phase (c) 180° out of phase (d) Independent of input				
3.	An ideal transformer is a transformer that has:	**(1)**	CO2	BL2	PO8
	(a) No losses (b) Copper loss only (c) Core loss only (d) Leakage flux only				
4.	If P = 800 W and S = 1000 VA, the power factor is:	**(1)**	CO1	BL1	PO1
	(a) 0.8 (b) 0.6 (c) 1 (d) 0.25				
5.	If voltage and current are in phase, the reactive power is:	**(1)**	CO1	BL1	PO1
	(a) Maximum (b) Negative (c) Infinite (d) Zero				
6.	The frequency of secondary voltage in a transformer is:	**(1)**	CO1	BL1	PO1

(a) Equal to the primary frequency (b) Half the primary frequency

(c) Double the primary frequency (d) Zero

7.	In an inverting amplifier, the input signal is applied to	**(1)**	CO3	BL3	PO3

(a) Inverting terminal (b) Non-inverting terminal

(c) Output terminal (d) Ground terminal

8.	The principle of operation of a transformer is based upon	**(1)**	CO3	BL3	PO1

(a) self induction (b) mutual induction

(c) static induction (d) dynamic induction

9.	In a fixed-bias transistor circuit, the base resistor is connected between	**(1)**	CO3	BL3	PO9

(a) Collector and Base (b) Base and Ground

(c) Base and Supply Voltage (d) Collector and Emitter

10.	The emitter-base junction of diode is usually	**(1)**	CO4	BL4	PO1

(a) Forward-biased (b) Reverse-biased

(c) Non-conducting (d) Operating in the breakdown region

11.	Form factor is defined as:	**(1)**	CO2	BL2	PO2

(a) Peak Value / RMS Value (b) RMS Value / Average Value

(c) Average Value / RMS Value (d) Peak Value / Average Value

12.	Which of the following is a semiconductor material?	**(1)**	CO4	BL3	PO9

(a) Copper (b) Silver

(c) Silicon (d) Aluminum

Part B
(Answer any three of the following) 3x5=15

13.	Define cycle, frequency, RMS, Form factor, Peak factor	**(5)**	CO2	BL3	PO2
14.	Write down the V-I characteristic of zenor diode.	**(5)**	CO3	BL4	PO3
15.	Write down the emf equation of DC machines	**(5)**	CO3	BL2	PO2
16.	Write down the voltage and current equation of ac wave form.	**(5)**	CO1	BL1	PO1
17.	Write down the working principle of pnp transistor	**(5)**	CO3	BL3	PO3

Part C

(Answer any three of the following) 3x15=45

18. Answer all

(a)	Write down the Pin configuration of OPAMP(IC741) with diagram.	**(7)**	CO2	BL1	PO12
(b)	What is an integrated Circuit? Write down the advantages, disadvantages	**(8)**	CO1	BL2	PO10

19. Answer all

(a)	What do you mean by Ideal transformer. Also draw the phasor diagram.	**(6)**	CO4	BL4	PO8
(b)	Draw and explain the equivalent diagram of transformer with referred to the secondary	**(9)**	CO4	BL5	PO8

20.	a reactor having negligible resistance and an inductance of 0.1H is connected in series with a resistor of 15Ohm. The circuit is connected across a 230V , 50Hz single phase ac supply. Find (i) currentflowing in the circuit (ii) power factor of the circuit (iii) voltage across reactor (iv) voltage across resistor	**(15)**	CO4	BL6	PO11
21.	Draw and explain Inverting & Non-Inverting Amplifier	**(15)**	CO1	BL2	PO10
22.	Calculate note current and mesh current using nodal analysis and mesh analysis for the given circuit where all resistance are 20 ohm and V1=50V and V2= 100V	**(15)**	CO3	BL5	PO4

